St. John College of Engineering and Management
(A Christian Religious Minority Institution)
Autonomous Institute
Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE
DTE Code : 3218 AICTE Permanent ID : 1-4790201
NAAC Accredited with Grade A+, Three Programs NBA Accredited

17 NOV 2025

End Semester Examination(SET E)

Course Name - Problem Solving Using C++ **Semester-I**

Course Code- 25FEES102 **Time-2hr30min**

Maximum Marks- 60

All Questions are Compulsory

Assume Suitable data wherever required

Q	Attempt all Questions	Marks Max	Course Outcome	RBT Level

Q1. a	Find smallest among three numbers.
	`#include <stdio.h>`
	`void main()`
	`{`
	`int a=4, b=9, c=2;`
	`if(a<b && a<c);`
	`printf("Smallest = %d", a);`
	`elseif(b<c || b<a)`
	`printf("Smallest = %d", b);`
	`else`
	`printf("Smallest = %d", c);`
	`}`
		5	1	L2

	or			

Q1.b	Check whether a number is positive or negative. Fix the error
	`#include <stdio.h>`
	`void main()`
	`{`
	`int n;`
	`printf("Enter number: ");`
	`scanf("%d", n);`
	`if(n>0);`
	`printf("Positive");`
	`else;`
	`printf("Negative");`
	`}`
		5	1	L2

Q2. a	Check if a number is even or odd.Fix the error
	`#include <iostream>`
	`using namespace std;`
	`int main()`
	`{`
	`int n=11`
	`if(n %2=0)`
	`cout<<"Even";`
	`else`
	`cout<<"Odd"`
	`}`
		5	2	L2

	or			

Q2. b	Swap two double numbers. Debug the code
	`#include <iostream>`
	`using namespace std;`
	`int main()`
	`{`
	`double x=3.5, y=8.5, temp;`
	`temp = x`
	`x= y`
	`y= temp`
	`cout<< x <<" "<< y;`
	`}`
		5	2	L2

Q3. a	Write a recursive function to calculate power (i.e., x^n).	15	3	L2
	Or			
Q3. b	Write a program in C++ to check whether a matrix is symmetric or not.	15	3	L2
Q4. a	Write a program in C++ to demonstrate a nested structure for storing employee details (Name, Department with dept_id and dept_name).	15	4	L3
	or			
Q4. b	Write a program in C to swap two numbers using pointers.	15	4	L3
Q5. a	Write a C++ program to demonstrate single inheritance with base class Person and derived class students.	15	5	L3
	or			
Q5. b	Write a program in C++ to create a class **Book** with data members (title, author, price). Create objects and display book details.	15	5 5	L3
Q6. a	Find the logical error in exception program:	5	6	L2

```
#include <iostream>

using namespace std;

int main() {

   try {

      throw "Error";

   } catch (int e) {

      cout << "Int: " << e;

   }

}
```

or

Q6.b Debug the error in the code. | 5 | 6 | L2

```
#include <iostream>
#include <fstream>
using namespace std;
int main()
{
    fstream file("data.txt", ios::in | ios::app);
    if(!file) {
        cout << "File not found";
    }
    file << "New line added\n";
    file.close();
}
```